

Sunburn device: the

The Sunburn Flowchart, built from the real blocks of the program

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 lights where you are through a phone, and then flashes, beeps or taps your arm until you put built from 76 coloured blocks. Every one of them is on these sheets, with a note saying

Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block forever block runs this plan round and round, all day.

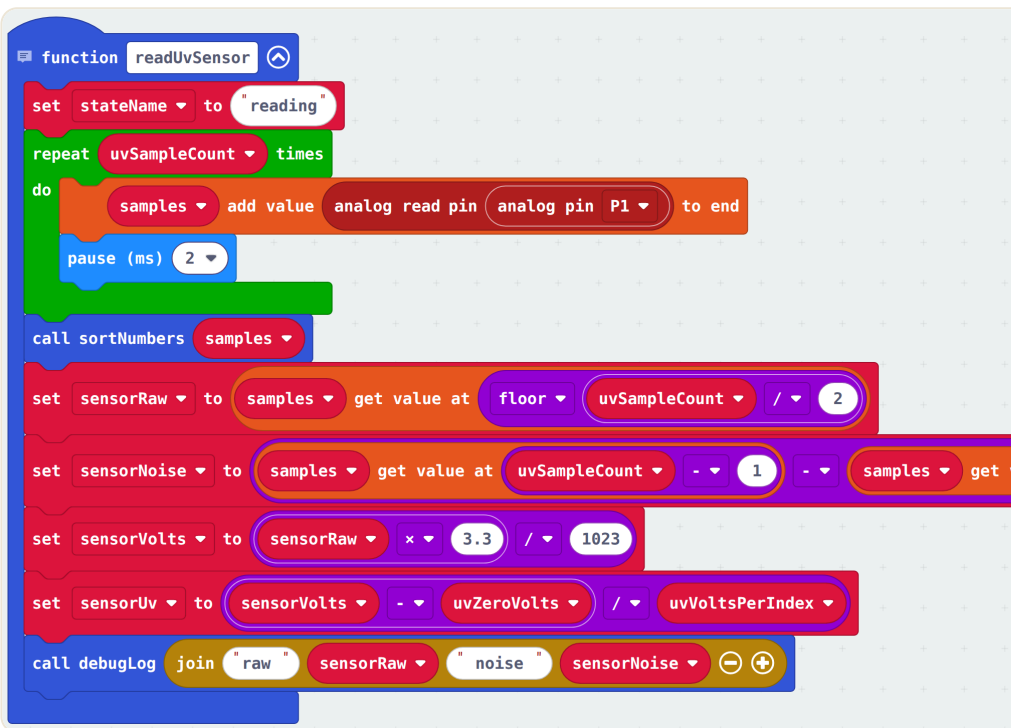
Device turns on



deviceTurnsOn

Switching on: shows a tick, says Device on on the screen, tells the app, and marks the device as on.

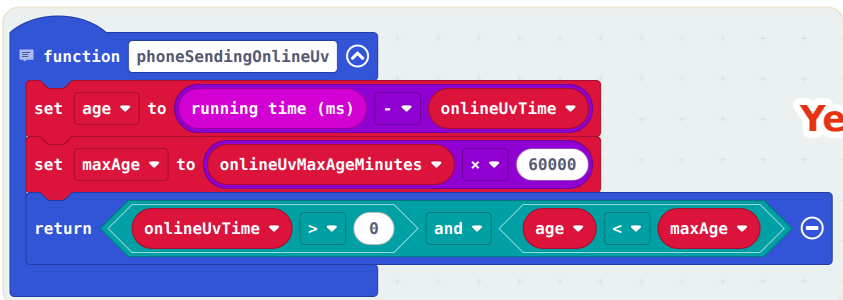
Read the UV sensor



readUvSensor

Reads the UV sensor plugged into pin P1. It takes 15 quick readings and uses the middle one, so a single odd reading cannot set off a false alarm. Then it turns the sensor's volts into a UV index number.

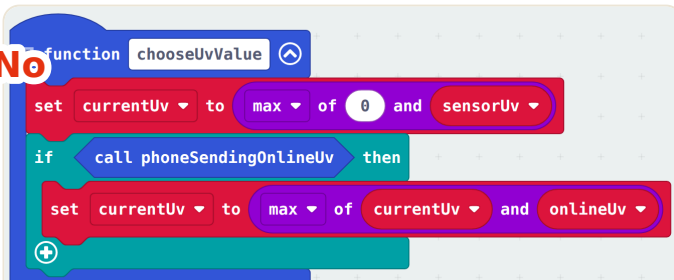
Is the phone sending the online UV index?



phoneSendingOnlineUv

Pick the UV value: the higher of sensor and online

Yes or No

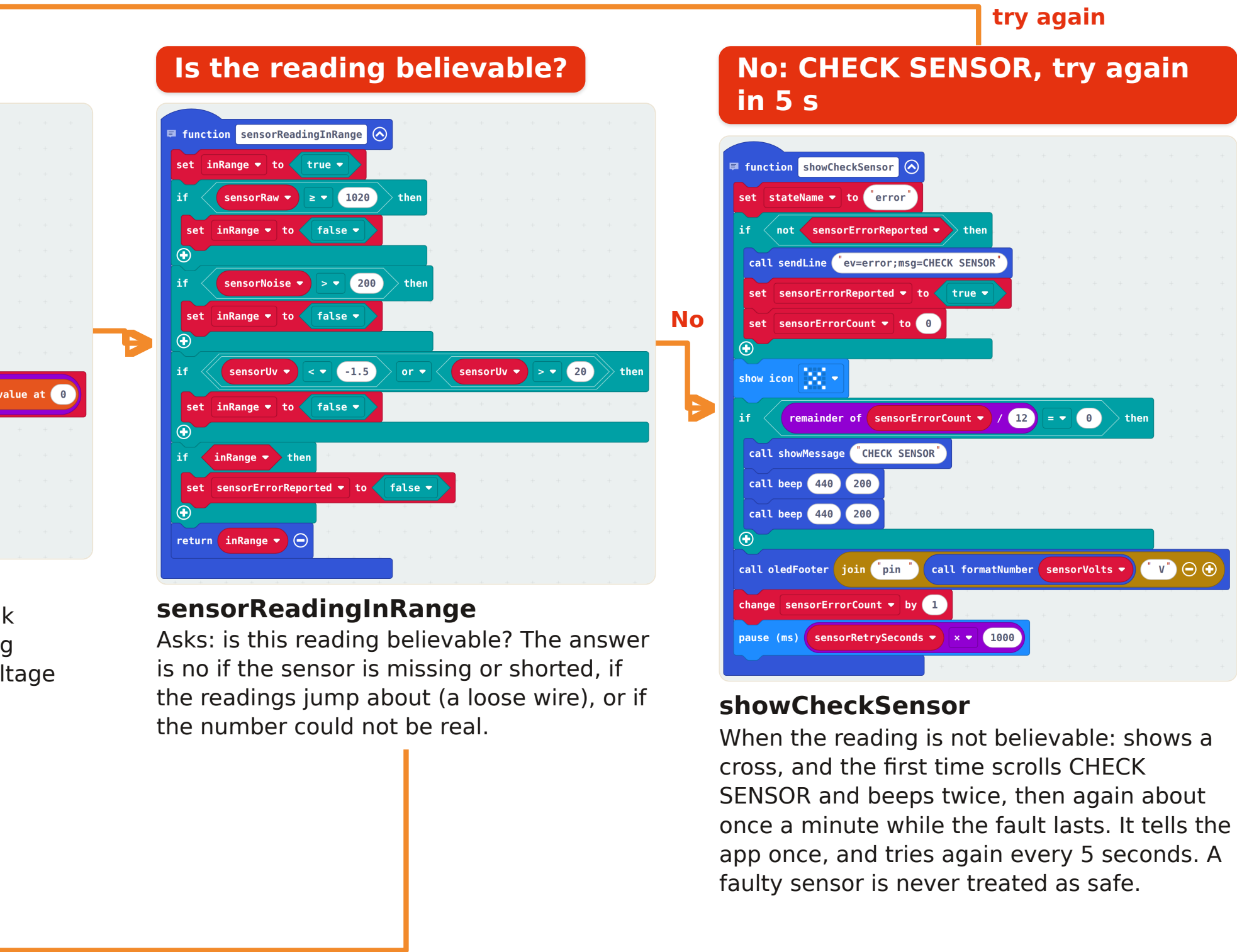


program on the wrist

am, with what each one does

hts and two buttons. It measures the sun's UV with a sensor, checks the online UV index for sunscreen on. Two hours later it reminds you to reapply. The program that does all this is what it does.

k of code that does that job, exactly as it looks in the MakeCode editor, and the note says what it does. The





The Sunburn Flowchart

phoneSendingOnlineUv
Asks: has the phone app sent an online UV index in the last 30 minutes?

chooseUvValue
Picks the UV number the device will act on: the higher of the sensor's reading and the phone's online value, or just the sensor's reading when the phone is not sending.

UV
Tu
fo
to
ve
ar

0 to 2

3 to 7

0 to 2, low: happy face

```
function showHappyFace
  set stateName to "safe"
  show icon
  call oledShowMessage "Low UV. No sunscreen needed"
```

showHappyFace
UV 0 to 2: shows a happy face. No sunscreen needed right now.

3 to 7, moderate or high
flash and slow beep un

```
function moderateOrHighUv
  if protectionLevel = 0 then
    set stateName to "alert"
    call showMessage "Sunscreen time"
    call flashAndBeepUntilA false
    if ackPressed then
      call showMessage "1 tsp SPF 50+ on each arm, each leg"
      call sunscreenApplied 1
    else if deviceOn then
      call standbyUntilButton
    else
      call alreadyProtected
```

moderateOrHighUv
UV 3 to 7: flashes and beeps slowly. If A is pressed, then shows how much sunscreen to put on and starts a timer. If sunscreen is already on, shows a tick and the time left. If A is pressed within 10 minutes, the device assumes it is not being worn and sleeps.

sunscreen already on A pressed

Sunscreen already on: just a tick

```
function alreadyProtected
  set stateName to "protected"
  show icon
  set timeLeft to reapplyDueTime - running time (ms)
  call oledShowMessage join "Sunscreen on. Reapply in " call formatTime timeLeft
```

alreadyProtected
Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

A pressed: sunscreen applied

```
function sunscreenApplied level
  show icon
  set protectionLevel to level
  change sunscreenCount by 1
  call sendEvent "sunscreen"
  call start2HourReapplyTimer
```

sunscreenApplied
A was pressed after alert: remembers what level of sunscreen was given, counts one more minute, tells the app, and starts the timer.

UV: back to the top

then back to the top

turns a UV index into one of the four levels on the flowchart: low (0 to 2), moderate or high (3 to 7), very high (8 to 10) or extreme (11 and above).

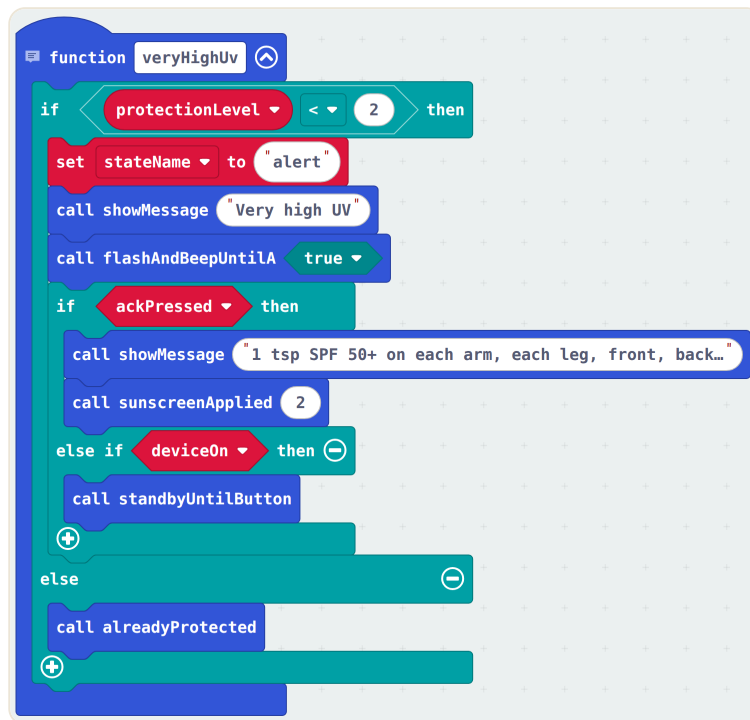
3 to 7

8 to 10

11 and above

High: Until A

8 to 10, very high: flash and fast beep until A

11
ta

veryHighUv

UV 8 to 10: flashes and beeps quickly until A is pressed, then shows the sunscreen advice plus hat and shade, and starts the 2 hour timer. It alerts again if the wearer so far only had the moderate advice.

wly until A
h
he 2 hour
it just
nobody
e device
d goes to

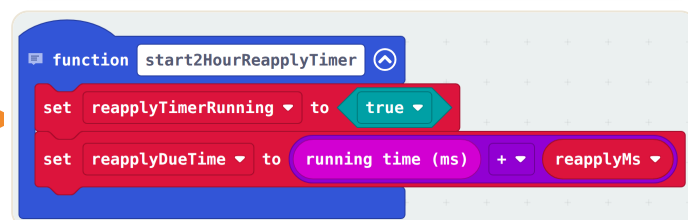
extr
UV 1
s pre
nste
sleep

A pressed

nscreen is

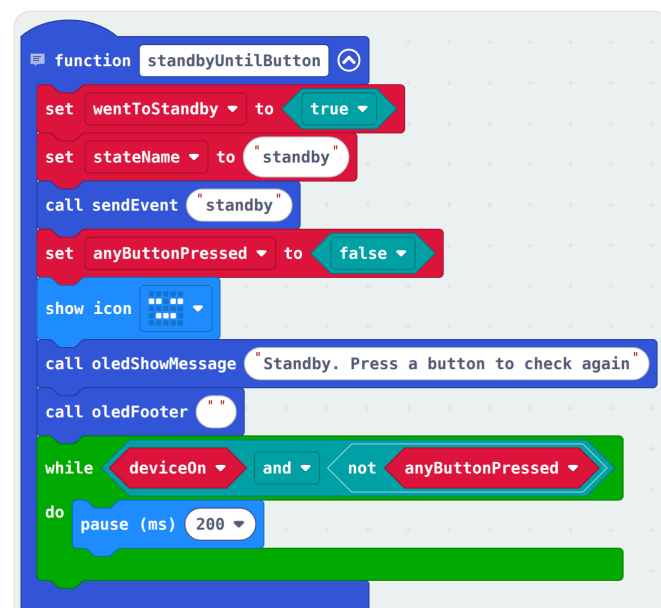
Start the 2 hour reapply timer

Sleep until any button is pressed



start2HourReapplyTimer

Starts the 2 hour countdown to the next sunscreen.

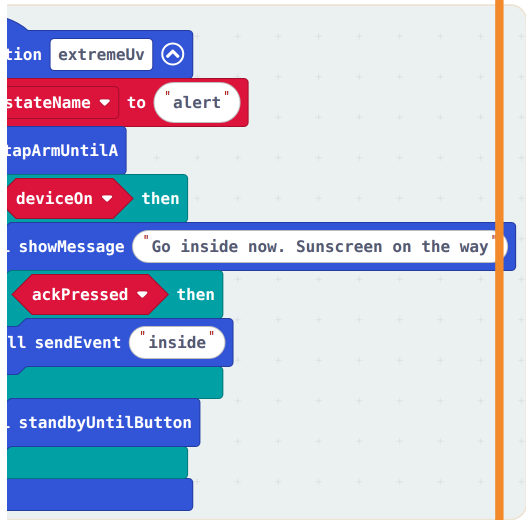


standbyUntilButton

Clean, melts for easy butter, or deep

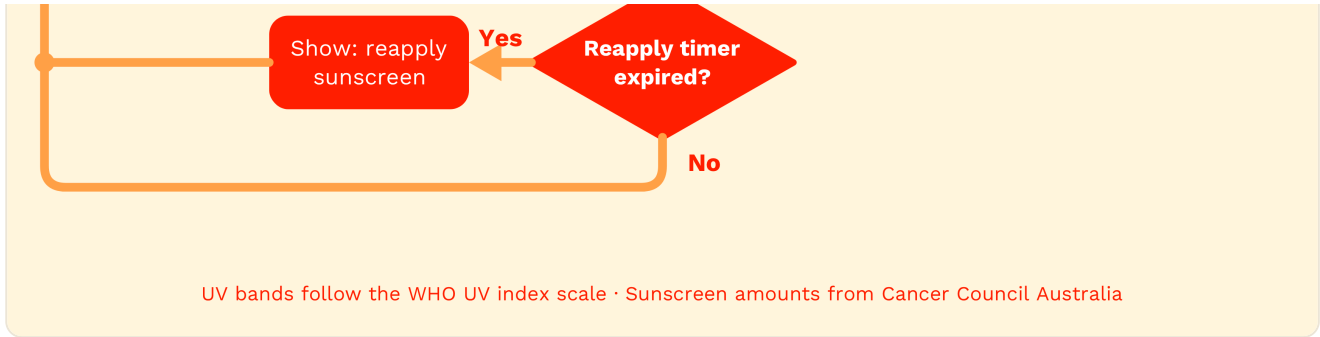
a button was pressed

and above, extreme:
p the arm until A



extremeUv

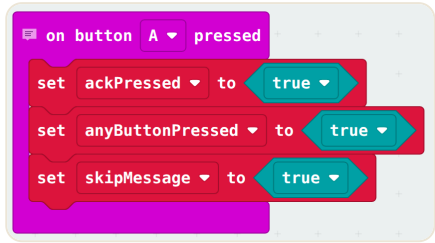
1 or more: taps the arm until A pressed (after 2 minutes it beeps loud), shows Go inside now, then os until a button is pressed.



The plan the device follows. Every box on it is a block on the left, with the same name.

The two buttons

A: done

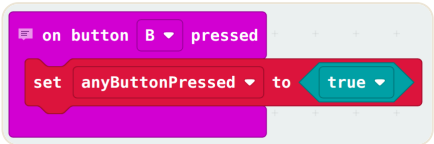


on button A pressed

Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

Pressing A and B together turns the device off, or on again (see the flowchart).

B: wake up



on button B pressed

Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.

The whole program



All 76 blocks in the editor, in nine sections. The flowchart on this sheet is sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth, maths and the optional screen.

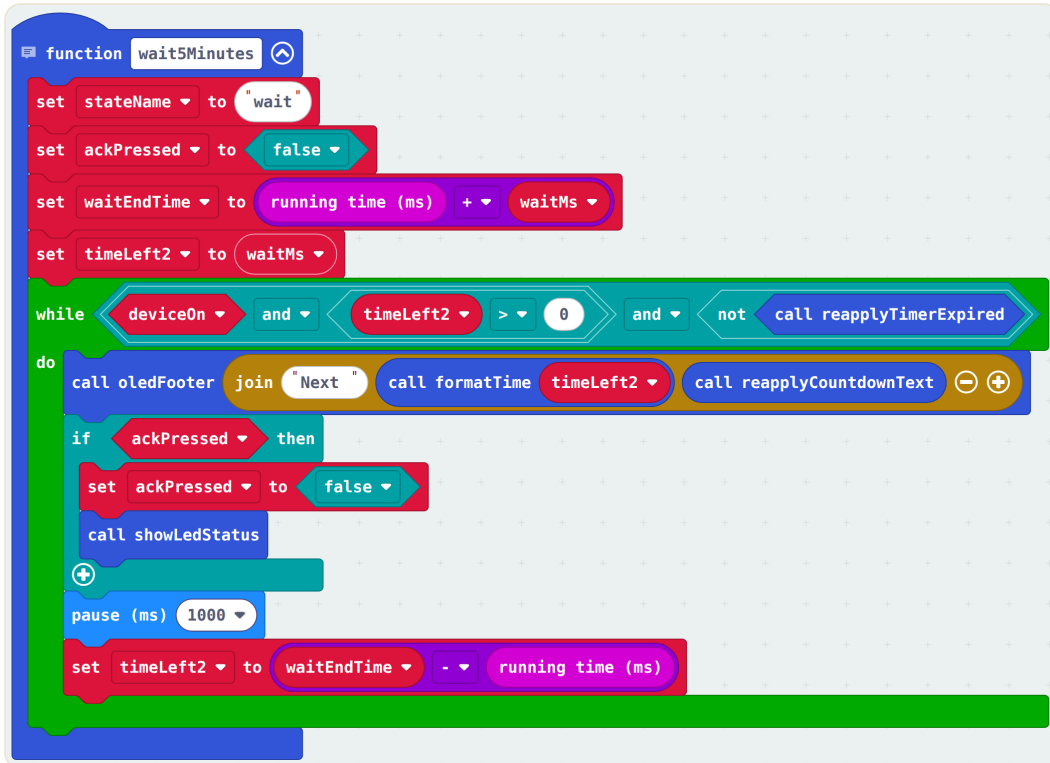
What the block colours mean

- | | |
|-------------------------------|-----------------------------|
| Variables: remember a value | Text |
| Functions: a step with a name | Arrays: lists |
| Basic: the lights, pauses | Music: beeps |
| Input: buttons and time | Pins: sensor, motor, screen |
| Loops: repeat | Bluetooth: the phone |
| Logic: if, compare | Serial: the USB cable |
| Maths | Control: timing |

Words used here

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.

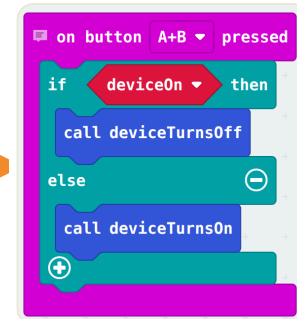
Wait 5 minutes



wait5Minutes

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

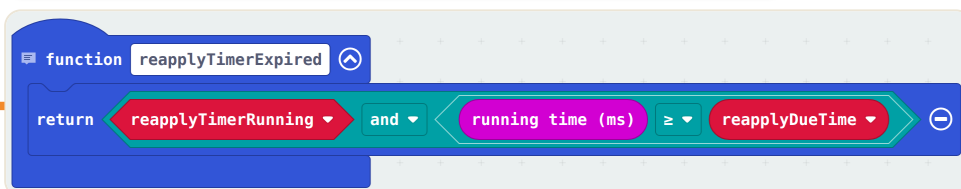
A and B held?



on button A+B pressed

Pressing A and B together turns the device off, or on again.

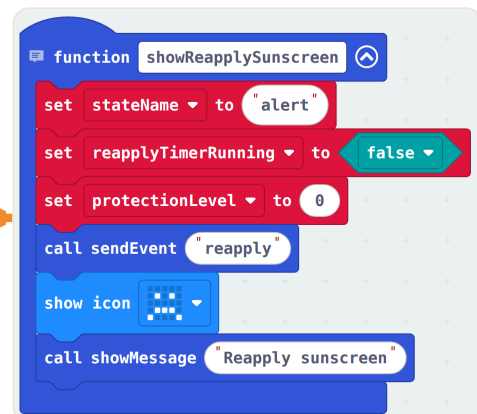
Is it time for more sunscreen?



reapplyTimerExpired

Asks: have the 2 hours since the last sunscreen passed?

Yes: show Reapply sunscreen



showReapplySunscreen

Shows Reapply sunscreen and clears the timer, so the next trip round the flowchart alerts again if the sun is still strong.

Sleep: waits for any button, or done from the app. Pressing A and B together still turns the device off. Afterwards it goes straight back to reading the sensor.

Yes: device turns off

```
function deviceTurnsOff
  set deviceOn to false
  set stateName to "off"
  call servoOff
  clear screen
  call oledClear
  call oledWriteRow 3 "OFF"
  call oledWriteRow 5 "A+B to turn on"
  call sendEvent "off"
```

deviceTurnsOff

Switching off: everything stops and the screen shows OFF. Bluetooth stays on so the app can switch it back on.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

OLED An optional small screen. Without one, the grid of lights does the job.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

Every one of the 76 blocks, with its note, is in the booklet on the table.